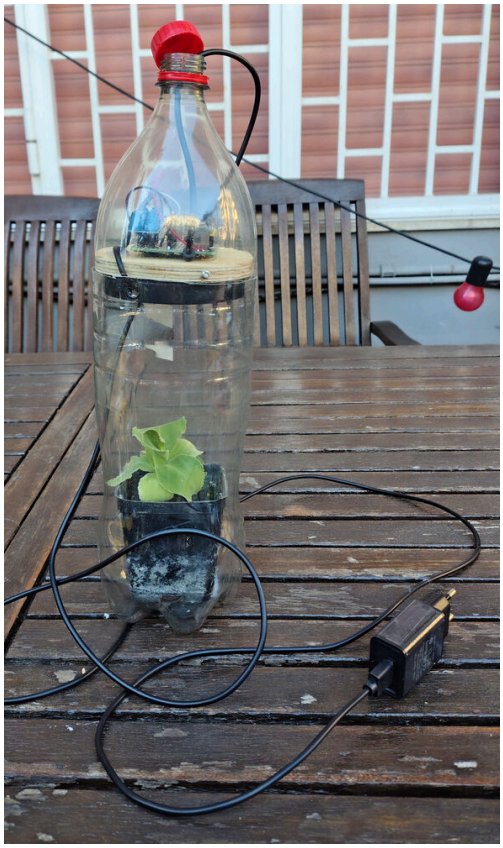
Build your SeedPeers micro greenhouse

A small greenhouse made from a recycled 2 L PET bottle. It has a grow light and an ESP32 that you control from your smartphone.



The finished micro greenhouse

> 🎬 **Video:** Watch the build

What you need

Materials

#	Item	Qty	Notes
1	Empty 2 L PET bottle with cap	1	Clear plastic, label removed
2	Plywood board	1	About __ mm thick (TODO). You cut a disc from it the size of the bottle
3	LED ring grow light	1	Outer diameter __ mm (TODO), __ V (TODO)
4	ESP32 dev board (ESP32-WROOM-32)	1	Mounted on a perfboard
5	Dual MOSFET driver module	1	Switches/dims the LED ring. TODO: model
6	Perfboard + jumper wires	1	
7	USB power supply + cable	1	5 V, ≥ __ A (TODO)
8	Small wood screws	4–6	To fix the board and the bottle top to the disc
9	Hot-glue sticks	–	
10	Small plant pot + soil + seedling	1	Must fit inside the bottle

Tools

- Cutter / sharp scissors
- Jigsaw (or coping saw)
- Hot-glue gun
- Screwdriver
- Pencil, masking tape
- Soldering iron *(optional, for the perfboard)*

> ⚠️ **Safety:** wear eye protection when you use the jigsaw. Keep fingers away from the blade and be careful with the cutter on curved plastic. Hot glue causes burns.

Step 1 – Cut the bottle



The bottle cut in two parts

1. Wrap a strip of masking tape around the bottle at __ **cm** from the base (TODO). Use it as your cutting guide.
2. Cut along the tape with the cutter, then finish the cut with scissors.
3. You now have two parts:
    - the **bottom part**, which holds the plant,
    - the **top part** (shoulder + cap), which becomes the lid.

Tip: smooth the cut edges so they don’t scratch.

Step 2 – Check the fit (basic greenhouse)



The basic greenhouse without electronics

Put the potted seedling in the bottom part and slide the top part over it. This is already a working passive greenhouse. The next steps add the light and the electronics.

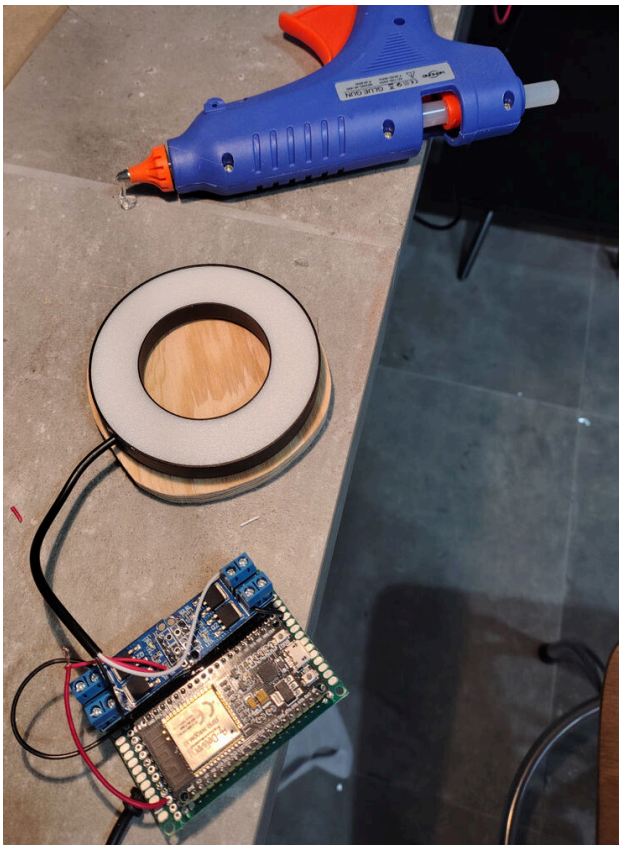
Step 3 – Cut the wooden disc



Tracing and cutting the disc with a jigsaw

1. Place the top part of the bottle on the plywood and trace around it with a pencil.
2. Cut out the disc with the jigsaw. Stay slightly **inside** the line so the disc fits snugly inside the bottle.
3. Sand the edges.

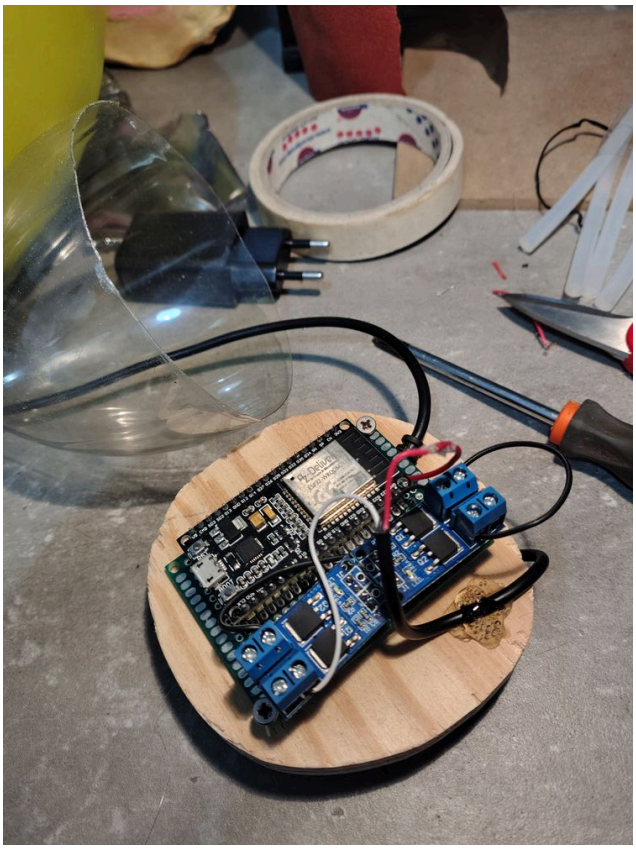
Step 4 – Mount the LED ring



LED ring glued to the disc, next to the electronics

1. Hot-glue the LED ring to the **underside** of the disc so the light points down at the plant.
2. Route the LED cable to the top side of the disc.

Step 5 – Mount the electronics



ESP32 and MOSFET module screwed to the disc

1. Screw the perfboard with the ESP32 and the MOSFET module onto the **top side** of the disc.
2. Wire the parts as follows:

From	To
ESP32 GPIO __ (TODO)	MOSFET module IN1
ESP32 GND	MOSFET module GND
LED ring + / –	MOSFET module OUT1
USB 5 V / GND	ESP32 VIN / GND and MOSFET module V+

3. Fix the cables with a drop of hot glue (strain relief).

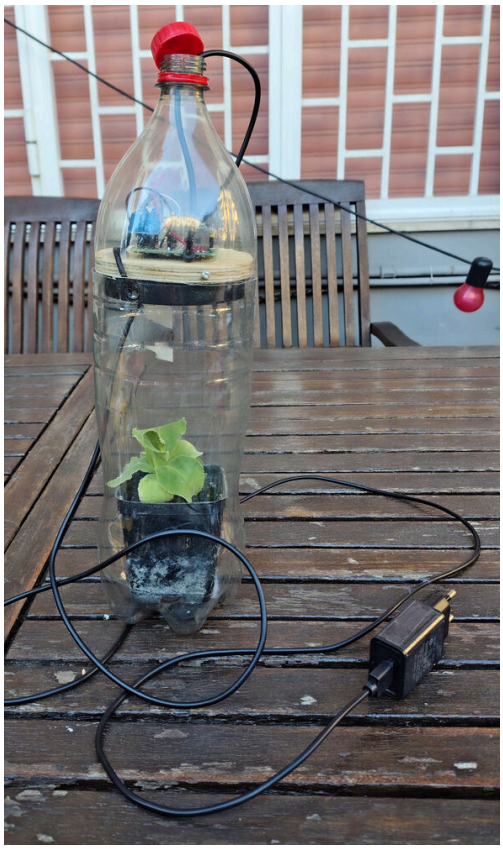
Step 6 – Close the lid



Top part of the bottle screwed onto the disc

1. Pass the power cable through the bottle neck. Drill a hole in the cap or leave the cap off.
2. Slide the top part of the bottle over the disc and fix it with small screws through the plastic into the edge of the disc.

Step 7 – Assemble the greenhouse



The fully assembled greenhouse

1. Put the potted plant in the bottom part.
2. Rest the lid unit (disc + electronics + top part) on the bottom part.
3. Plug the USB cable into the power supply.

Step 8 – Upload the firmware

1. Install the Arduino IDE (or PlatformIO) and the ESP32 board support.
2. Open the firmware: `TODO: link to repository / sketch`.
3. Set your parameters (TODO: Wi-Fi / Bluetooth, RemoteXY token, …).
4. Upload the sketch to the ESP32 over USB.

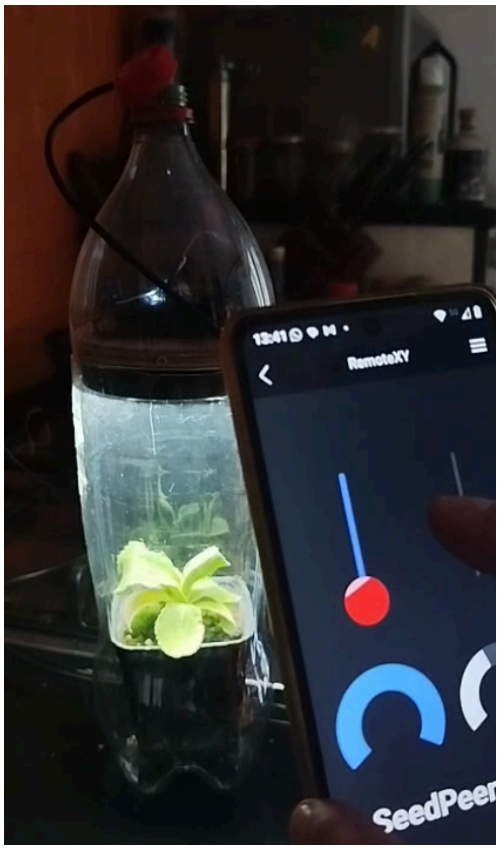
Step 9 – Turn on the grow light



Grow light on

When powered, the LED ring lights up the plant with the grow spectrum (pink/purple light).

Step 10 – Control it from your smartphone



Control from the RemoteXY app

1. Install **RemoteXY** on your phone (Android / iOS).
2. Connect to the **SeedPeers** device (TODO: connection method).
3. Use the controls to set the light intensity and TODO (other controls).

Troubleshooting

Problem	Possible cause	Fix
The LED ring doesn't turn on	Wiring / wrong GPIO	Check the step 5 wiring
The app can't find the device	Firmware not uploaded / wrong token	Repeat step 8
Condensation on the walls	Too much humidity	Remove the cap to let air in
TODO		

Next steps

- TODO: add sensors (temperature, humidity, soil moisture)

- TODO: share your data with the SeedPeers community